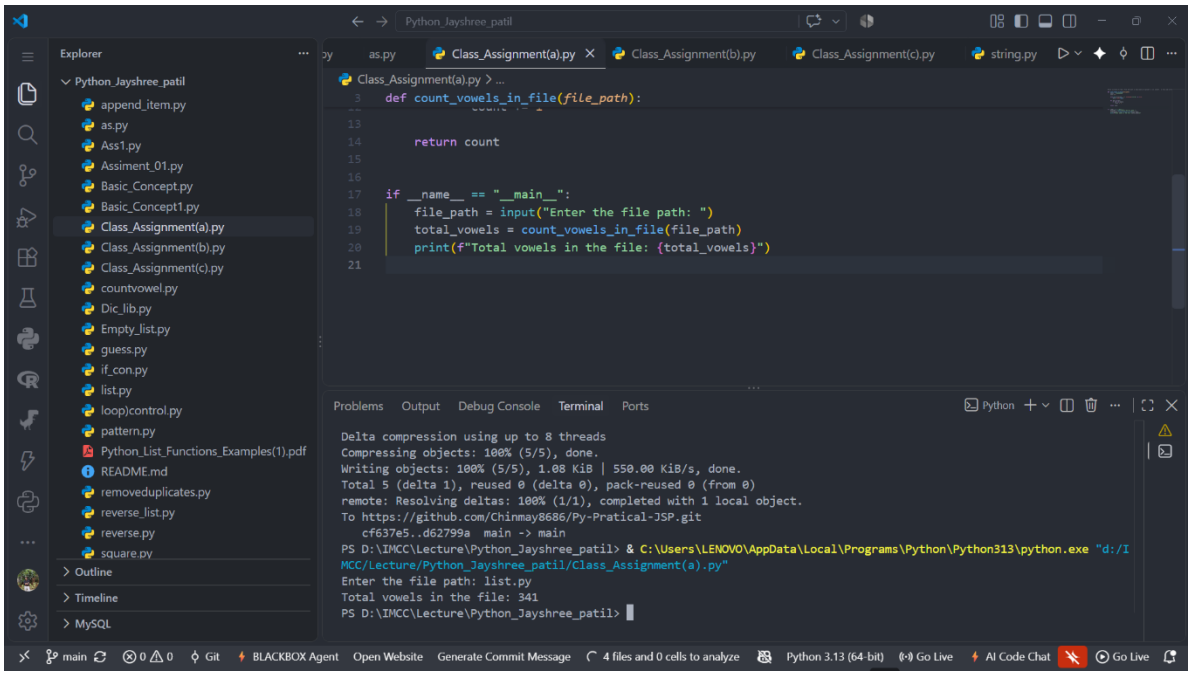
Academic Year 2026-27
FY MCA- SEM I
Class Assignment1

Roll No: 21601206
Name: Chinmay Yannawar

01] write a program to read a file and count ow many vowels are present in its content . Is this code correct ?

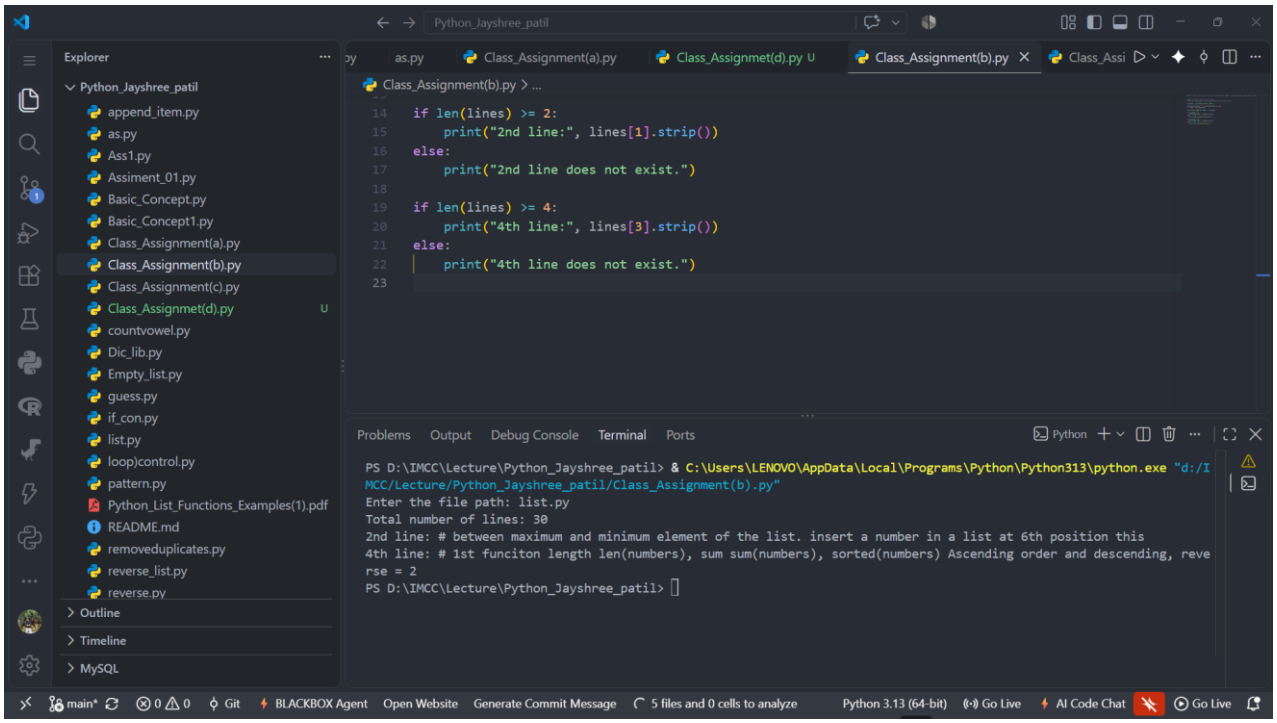
```
def count_vowels_in_file(file_path):
    vowels = "aeiouAEIOU"
    count = 0
    with open(file_path, "r", encoding="utf-8") as file:
        text = file.read()
    for char in text:
        if char in vowels:
            count += 1
    return count
if __name__ == "__main__":
    file_path = input("Enter the file path: ")
    total_vowels = count_vowels_in_file(file_path)
    print(f"Total vowels in the file: {total_vowels}")
```



02] Read all lines from a file into a list. print the total number of lines and display te 2nd and 4 th line.

Solution :

```
file_path = input("Enter the file path: ")
with open(file_path, "r", encoding="utf-8") as file:
    lines = file.readlines()
print("Total number of lines:", len(lines))
if len(lines) >= 2:
    print("2nd line:", lines[1].strip())
else:
    print("2nd line does not exist.")
if len(lines) >= 4:
    print("4th line:", lines[3].strip())
else:
    print("4th line does not exist.")
```



03. Take a sentence from the user, split it inot words, abd store them in a list. print te longest word.

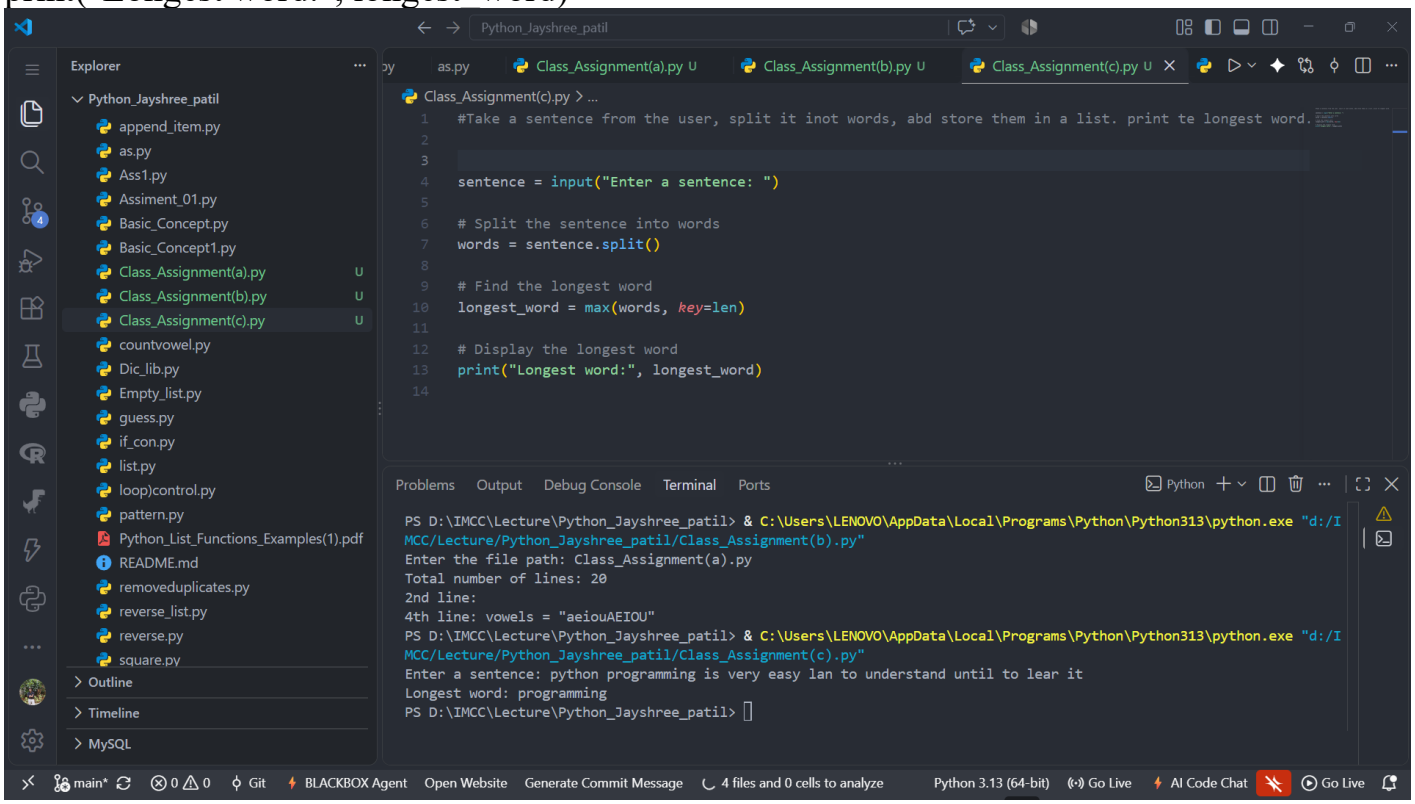
Solution:

```
sentence = input("Enter a sentence: ")

# Split the sentence into words
words = sentence.split()

# Find the longest word
longest_word = max(words, key=len)

# Display the longest word
print("Longest word:", longest_word)
```



04] Read a file, split its content into words, and store them in a list.

i. Print the total number of words

ii. Print all unique words

Solution:

```
# Read a file, split its content into words, and store them in a list.

file_path = input("Enter the file path: ")

with open(file_path, "r", encoding="utf-8") as file:
    text = file.read()

words = text.split()

# i. Print the total number of words
print("Total number of words:", len(words))

# ii. Print all unique words
unique_words = set(words)
print("Unique words:", unique_words)
```

